Session 11: Linux Networking Lab

Exercise 1 — Discover Your Network Interfaces

Commands

- nmcli
- ifconfig
- ip addr

Questions

1. Which interface is your primary network interface?
2. Does your primary interface have an IPv4 address? Verify using ip addr show
3. Which IPv6 addresses are assigned?
4. What is the status of your device and its active connection ?

Answers

1. Primary interface
Command: ***nmcli device show***
Explanation: Shows interface state; look for "connected".
Example output:

```
GENERAL.DEVICE:        enp0s3
GENERAL.TYPE:          ethernet
GENERAL.HWADDR:        08:00:27:BD:17:CD
GENERAL.MTU:           1500
GENERAL.STATE:         100 (connecté)
```

2. IPv4 assigned?
Command: ***ip addr show enp0s3***
Explanation: Look for inet.
Example:

inet 10.0.2.15/24

3. IPv6 assigned?
Command: ***ip addr show enp0s3***

Explanation: Look for inet6.
Example:

inet6 fd17:625c:f037:2:8d2c:6853:c89f:4785/64

4. Device & connection status
Command(s):

- nmcli device status
- nmcli connection show
  Explanation: Lists device states and active profile.
  Example:

enp0s3 connected Wired connection 1

NAME	UUID	TYPE	DEVICE
Connexion filaire 1	fb1467c6-62a3-39e4-be6e-d8bd9591a477	ethernet	enp0s3
lo	599387ab-020d-45e5-ac2d-2f012c41c908	loopback	lo

Exercise 2 — Disable and Enable a Network Interface (15 min)

Commands

- ip link set
- ping google.com

Questions

1. Disable your network interface and check using ping (e.g. google.com). What happens?
2. What happens when the interface is up again?

Answers

1. Interface down
Command: ***sudo ip link set enp0s3 down and ping google.com***
Explanation: Interface is down → no connectivity.
Example:ping 8.8.8.8
ping: connect: Le réseau n'est pas accessible

2. Interface up
Command: ***sudo ip link set enp0s3 up and ping google.com***
Example:

PING google.com (142.250.74.238) 56(84) bytes of data.
64 bytes from par10s40-in-f14.1e100.net (142.250.74.238): icmp_seq=1 ttl=255 time=10.9 ms

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Exercise 3 — Assign a Static IPv4 Address

Your network interface is enp0s3. Assign the following network settings:

- **Static IP:** 172.25.250.11
- **Subnet mask:** 255.255.255.0 (/24)
- **Gateway:** 172.25.250.254
- **DNS server:** 172.25.250.254
- **Connection name:** lab

Your task is to configure your interface with these settings using NetworkManager (nmcli), then verify it.

Answers

Command: ***nmcli con add con-name lab ifname enp0s3 type ethernet ipv4.method manual ipv4.addresses 172.25.250.11/24 ipv4.gateway 172.25.250.254 ipv4.dns 172.25.250.254***

Activate the connection: ***nmcli con up lab***

Check the ip ***ip addr show enp0s3***

Check network details ***nmcli device show enp0s3***

Exercise 4 — Add a Second IPv4 Address

Assign a second Ipv4 address (10.0.1.1/24) to your “lab” connection and verify that both IPs are on your interface. Why do you thing it is useful to have multiple IP.

Answers

Add IP: Command : ***nmcli con mod lab +ipv4.addresses 10.0.1.1/24*** and ***nmcli con up lab***

Check
Command: ***ip addr show enp0s3***
Explanation: Needed for multi-subnet hosting, lab testing, routing tests.

Exercise 5 — Assign an IPv6 Address (15 min)

Add IPv6 address (*2001:db8::1/64) to your interface (e.g. enp0s3), for this use ip*

1. Which command assigns the IPv6 address?
2. Which command verifies it?
3. Which IPv6 addresses appear?

Answers

1. Add IPv6 address
Command:

sudo ip -6 addr add 2001:db8::1/64 dev enp0s3

2. Verify
Command:

ip -6 addr show enp0s3

3. Example output

```
inet6 2001:db8::1/64 scope global
   valid_lft forever preferred_lft forever
```

Exercise 6 — Explore Routing

Commands

- ip route

Questions

1. Which interface is used to reach the Internet? Try using google.com
2. How do you remove the temporary route?

Answers

1. Interface used for external traffic
Command: ***ip route get 8.8.8.8***
Example:

```
8.8.8.8 via 172.25.250.254 dev enp0s3 src 172.25.250.11 uid 1000
    cache
```

Command: ***ip route show***
Example:

```
default via 172.25.250.254 dev enp0s3 proto static metric 100
10.0.1.0/24 dev enp0s3 proto kernel scope link src 10.0.1.1 metric 100
172.25.250.0/24 dev enp0s3 proto kernel scope link src 172.25.250.11 metric 100
```

2. Remove route
Command:

sudo ip route del default via 172.25.250.254

Exercise 7 — Configure /etc/hosts

Assign a custom hostname `private` to the IP address `10.0.1.1`.

Verify that your computer can resolve the hostname and communicate with it using `ping`.

Why?

As seen in class, normally, your computer uses DNS to convert hostnames (like google.com) into IP addresses.

/etc/hosts is a local file that lets your system resolve hostnames without using DNS

Any command using private (like ping private or ssh private) will directly use 10.0.1.1.

This is very fast and works even if DNS is not available

Answers

Add a line mapping the IP address to the hostname

sudo sh -c 'echo "10.0.1.1 private" >> /etc/hosts'

Verify:

ping -c 3 private

Exercise 8 — Explore /etc/services (5 min)

Questions

1. What ports correspond to SSH and HTTP?

Answers

Command: ***grep -i ssh /etc/services***
Example: ssh 22/tcp

Command: grep -i http /etc/services
Example: http 80/tcp

Exercise 9 — Create a Virtual Interface

Create a virtual network interface , assign an IP, bring it up, and verify its configuration.

Commands

- sudo ip link add link DEVICE name macvtap1 type macvtap mode bridge
- ip addr
- sudo ip link set
- ip addr show

Answers

sudo ip link add link enp0s3 name macvtap1 type macvtap mode bridge

sudo ip addr add 172.16.99.100/24 dev macvtap1

sudo ip link set macvtap1 up

ip addr show macvtap1

Exercise 10 — Troubleshoot Connectivity

Questions

1. Is your gateway reachable?
2. Which ports are listening?

Answers

1. Gateway reachable
Command: ping -c 3 172.25.250.11 (the address seen when you do ip addr show)

2. Listening ports
Command: ss -tuln
Explanation: Shows TCP/UDP sockets.